

ELECTROCOOKIE LARGE · 30 × 10 + 4 RAILS · CONSOLIDATED BUILD

Lake node build map

Everything from the working prototype on one board, plus the one circuit that has never been built: a GPIO-controlled high-side switch so the boost converter and pressure loop only draw current during a reading. Twenty jumpers, ten discretes, six modules, nine nets — every hole verified against the net list. Discrete values are set for what the Onion Omega2 Maker Kit already provides; only the MOSFET and the shunt have to be bought.

JUMPERS	DISCRETES	NETS	LOOP SUPPLY
20	10	9	24 V

I²C ADDRESS

0x48

BOARD LAYOUT

All nets

3V3

GND

SW 3V3

24 V

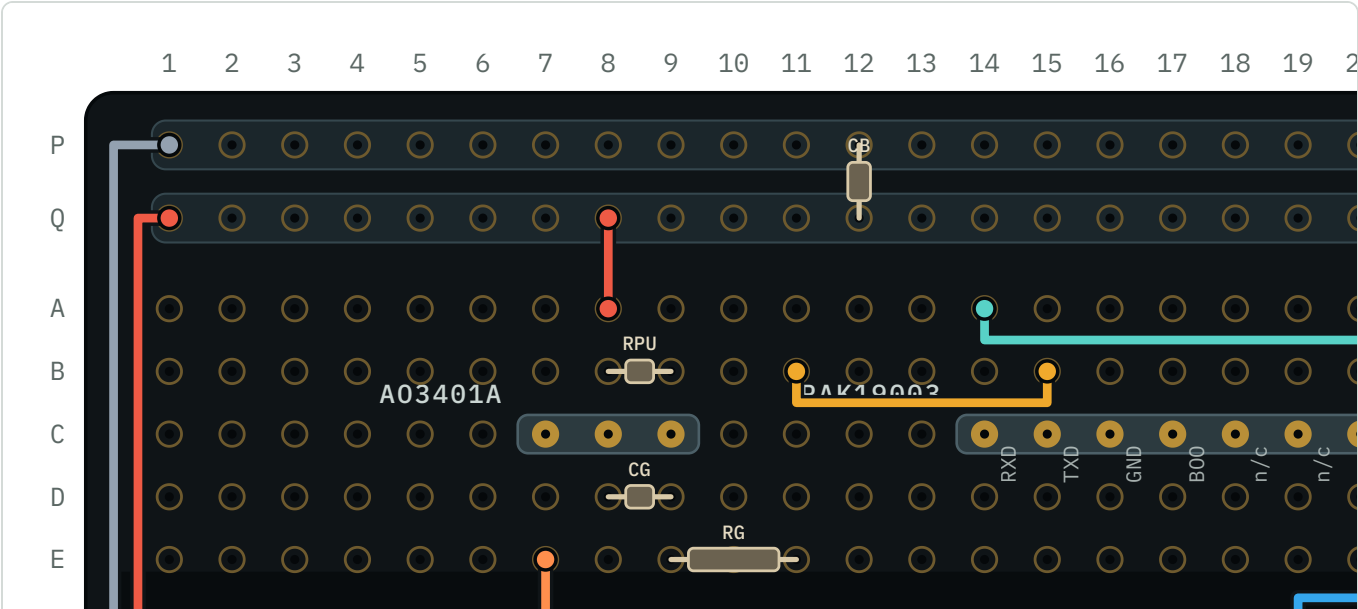
A0 sense

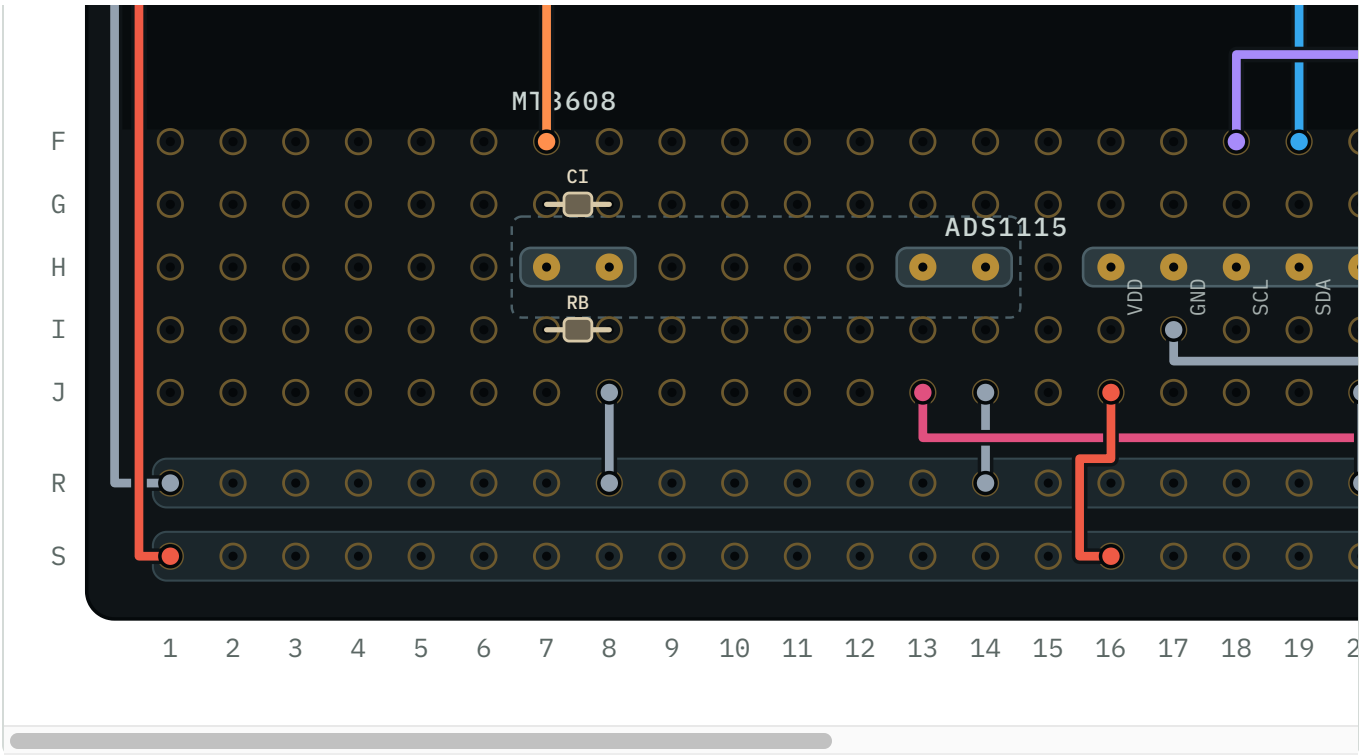
SDA

SCL

FET gate

1-Wire





The floorplan does the electrical work. The MT3608 sits at columns 7-14 and the shunt at 28-30, nearly two inches apart, so the switcher's 1.2 MHz input current never shares copper with the measurement. Modules sit in rows C and H rather than hard against the trench, which leaves the outer rows free for rail risers and the inner rows free for trench crossings. Wires drawn offset into the gaps between rows are point-to-point runs — the offset only keeps them from reading as connections to holes they pass over.

SOLDER ORDER

0 / 20 soldered CLEAR

RAILS AND POWER ENTRY

- 01

☐

P1 → R1

PRECUT

Ground rails joined — round the left end of the board
- 02

☐

Q1 → S1

PRECUT

3V3 rails joined — round the left end, outside the ground link
- 03

☐

A24 → Q24

PRECUT

RAK VDD (C24) straight up column 24 to the 3V3 rail
- 04

☐

A23 → P23

PRECUT

RAK GND (C23) to the outer ground rail, hopping the 3V3 rail

MOSFET SWITCH — THE NEW BLOCK

05

☐

A8 → Q8
PRECUT AO3401A source (C8) up to always-on 3V3, clear of RPU and CG

06

☐

B15 → B11
PRECUT RAK TXD (C15) to the gate resistor

07

☐

E7 → F7
PRECUT AO3401A drain (C7) straight down column 7 to the boost input

BOOST CONVERTER

08

☐

J8 → R8
PRECUT MT3608 IN– down to the ground rail — the high-current return

09

☐

J14 → R14
PRECUT MT3608 OUT– down to the ground rail

10

☐

J13 → J27
PRECUT 24 V from MT3608 OUT+ to the transmitter JST

MEASUREMENT

11

☐

I30 → I17
PRECUT KELVIN SENSE: shunt low side to the ADS GND pin — carries no loop current

12

☐

J30 → R30
PRECUT Loop return: shunt low side down to the ground rail

13

☐

J26 → J22
PRECUT Filtered sense to ADS A0 (H22)

14

☐

J16 → S16
PRECUT ADS VDD (H16) down to the 3V3 rail, hopping the ground rail

15

☐

J20 → R20
PRECUT ADS ADDR (H20) pulled low — address 0x48

SENSOR CONNECTORS AND DIGITAL

16

☐

E21 → F19
FORMED SDA — RAK C21 across the trench to ADS H19

- 17

☐

E22 → F18

FORMED

SCL — RAK C22 across the trench to ADS H18
- 18

☐

A27 → A14

PRECUT

DS18B20 data to RAK RXD (C14)
- 19

☐

A26 → Q26

PRECUT

DS18B20 + up to the 3V3 rail
- 20

☐

A28 → P28

PRECUT

DS18B20 – up to the ground rail, hopping the 3V3 rail

Do the under-module jumpers first. The RAK body overhangs rows A-D across columns 17-27 and the ADS body rows G-J across 16-25, with the RAK in row C and the ADS in row H, every jumper now leaves its half-row from the hole nearest its destination — nothing runs over another lead on the same node. A few still pass under a module body: B15, A23, A24 and the 1-Wire run under the RAK; I17, I20, I22, J16, J20 and J22 under the ADS. Run those first.

DISCRETES

REF	VALUE	FUNCTION	HOLES
RG	5.1k	gate series	E9 ↔ E11
RPU	51k	gate pull-up	B9 ↔ B8
CG	100nF	gate-source	D9 ↔ D8
CB	DNF	optional bulk, see note	Q12 ↔ P12
CI	100nF	boost bypass	G7 ↔ G8
RB	5.1k	switched-rail bleed	I7 ↔ I8
RSH	100 Ω	shunt, 1% metal film	H28 ↔ H30
RF	1k	A0 filter	I28 ↔ I26
CF	100nF	A0 filter at the ADC	I22 ↔ I20
R1W	5 1k	1-wire null-in	R27 ↔ R26

Values are set for the Onion Maker Kit. RG 5.1 k with RPU 51 k gives $V_{GS} = -3.0$ V, and $RG \times CG = 464$ μ s of gate ramp. With CI as a 100 nF bypass — the MT3608 carries its own input cap — that puts turn-on inrush around 50 mA, which is why CB is **DNF**: there is no longer anything for a bulk cap to buffer. The footprint stays for the case where the rail visibly sags.

WHY THE GROUND IS WIRED THIS WAY

DAISY-CHAINED – 20 mA through the reference



drop between shunt and ADS GND lands in the reading

KELVIN – reference carries nothing



Both ends of the shunt reach the ADS as a matched pair: A0 through the filter, ground through a wire carrying nothing but the ADS's own microamps. The loop's 20 mA leaves for the rail from the shunt itself, so its IR drop is outside the measurement entirely.